Standard Specification for

Corrugated Steel Pipe, Metallic-Coated, for Sewers and Drains

AASHTO Designation: M 36-03 (2007)[1]
ASTM Designation: A 760/A 760M-01a

American Association of State Highway and Transportation Officials
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1. SCOPE

1.1. This specification covers corrugated steel pipe intended for use for storm water drainage, underdrains, the construction of culverts, and similar uses. Pipe covered by this specification is not normally used for the conveyance of sanitary or industrial wastes. The steel sheet used in fabrication of the pipe has a protective metallic coating of zinc (galvanizing), aluminum Type 2, 55 percent aluminum-zinc alloy, zinc-5 percent aluminum-mischmetal alloy, or aluminum Type 1.

1.1.1. Steel sheet with zinc and aramid fiber composite coating may be specified for fabrication of pipe. Pipe made from sheet with this composite coating is always furnished with an asphalt coating. Therefore, the requirements in this specification should be considered as applying to a semi-finished pipe; the finished pipe must include provisions of M 190.

Note 1—Pipe fabricated with zinc and aramid fiber composite coated sheet and asphalt post-coating may be used for sanitary sewers and industrial applications. Petroleum products or similar materials in the sewer effluent may affect the performance of the asphalt coating.

1.2. The several different metallic coatings may not provide equal protection of the base metal against corrosion or abrasion in all environments, or both. Some environments may be so severe that none of the metallic coatings included in this specification will provide adequate protection. Additional protection for corrugated steel pipe can be provided by use of coatings applied after fabrication of the pipe as described in M 190, or by use of polymer precoated corrugated steel pipe as described in M 245.

1.3. This specification does not include requirements for bedding, backfill, or the relationship between earth cover load and sheet thickness of the pipe. Experience has shown that the successful performance of this product depends upon the proper selection of sheet thickness, type of bedding and backfill, controlled manufacture in the plant, and care in the installation. The installation procedure is described in AASHTO's *Standard Specifications for Highway Bridges*, Division II, Section 26.

2. REFERENCED DOCUMENTS

2.1. *AASHTO Standards*:

- M 190, Bituminous-Coated Corrugated Metal Culvert Pipe and Pipe Arches
- M 218, Steel Sheet, Zinc-Coated (Galvanized), for Corrugated Steel Pipe
- M 232M/M 232, Zinc Coating (Hot-Dip) on Iron and Steel Hardware
- M 245, Corrugated Steel Pipe, Polymer-Precoated, for Sewers and Drains
- M 274, Steel Sheet, Aluminum-Coated (Type 2), for Corrugated Steel Pipe

- M 289, Aluminum-Zinc Alloy Coated Sheet Steel for Corrugated Steel Pipe
- M 291M, Carbon and Alloy Steel Nuts [Metric]
- M 298, Coatings of Zinc Mechanically Deposited on Iron and Steel
- M 315M, Joints for Circular Concrete Sewer and Culvert Pipe, Using Rubber Gaskets [Metric]
- T 65M/T 65, Mass [Weight] of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings
- T 213M/T 213, Mass [Weight] of Coating on Aluminum-Coated Iron or Steel Articles
- T 241, Helical Continuously Welded Seam Corrugated Steel Pipe
- T 249, Helical Lock Seam Corrugated Pipe
- *Standard Specifications for Highway Bridges*

2.2. *ASTM Standards*:

- A 780, Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
- A 796/A 796M, Structural Design of Corrugated Steel Pipe, Pipe-Arches, and Arches for Storm and Sanitary Sewers and Other Buried Applications
- A 885, Steel Sheet, Zinc and Aramid Fiber Composite Coated for Corrugated Steel Sewer, Culvert, and Underdrain Pipe
- A 929/A 929M, Steel Sheet, Metallic-Coated by the Hot-Dip Process for Corrugated Steel Pipe
- B 633, Electrodeposited Coatings of Zinc on Iron and Steel
- D 1056, Flexible Cellular Materials—Sponge or Expanded Rubber
- F 568M, Carbon and Alloy Steel Externally Threaded Metric Fasteners

3. TERMINOLOGY

3.1. *Description of Terms Specific to This Standard*:

3.1.1. *fabricator*—the producer of the pipe.

3.1.2. *manufacturer*—the producer of the sheet.

3.1.3. *minimized coating structure*—a coating characterized by a finer metallurgical coating structure obtained by a treatment designed to restrict the formation of the normal coarse-grain structure formed during solidification of the Zn-5 Al-MM alloy coating.

3.1.4. *purchaser*—the purchaser of the finished product.

3.1.5. *regular coating structure*—the normal coating structure resulting from unrestricted grain growth during normal solidification of the Zn-5 A1-MM alloy coating.

3.2. *Abbreviations*:

3.2.1. *55 Al-Zn*—55 percent aluminum-zinc.

3.2.2. *MM*—mischmetal,

3.2.3. *Zn-5 Al-MM*—zinc-5 percent aluminum-mischmetal, 3.2.4. *AlT2*—aluminum-coated Type 2, and

3.2.5. *AlT1*—aluminum-coated Type 1.

4. PIPE CLASSIFICATION

4.1. The corrugated steel pipe covered by this specification is classified as follows:

4.1.1. *Type I*—This pipe shall have a full circular cross section, with a single thickness of corrugated sheet, fabricated with annular (circumferential) or helical corrugations.

4.1.2. *Type IA*—This pipe shall have a full circular cross section, with an outer shell of corrugated sheet and an inner liner of smooth (uncorrugated) sheet, fabricated with helical corrugations and lock seams.

4.1.3. *Type IR*—This pipe shall have a full circular cross section with a single thickness of smooth sheet, fabricated with helical ribs projecting outwardly.

4.1.4. *Type II*—This pipe shall be a Type I pipe which has been reformed into a pipe-arch having an approximately flat bottom.

4.1.5. *Type IIA*—This pipe shall be a Type IA pipe, which has been reformed into a pipe-arch having an approximately flat bottom.

4.1.6. *Type IIR*—This pipe shall be a Type IR pipe which has been reformed into a pipe-arch having an approximately flat bottom.

4.1.7. *Type III*—This pipe, intended for use as underdrains or for underground disposal of water, shall be a Type I pipe which has been perforated to permit the inflow or outflow of water.

4.1.8. *Type IIIA*—This pipe, intended for use as underdrains, shall consist of a semicircular cross section having a smooth (uncorrugated) bottom with a corrugated top shield.

4.2. Perforations in Type III pipe are included in three classes as described in Section 8.3.2.

4.3. Zn-5 Al-MM alloy-coated material is available in two coating classes, or structures, as follows:

4.3.1. *Class A*—Minimized coating structure.

4.3.2. *Class B*—Regular coating structure.

5. ORDERING INFORMATION

5.1. Orders for material to this specification shall include the following information as necessary, to adequately describe the desired product:

5.1.1. Name of material (corrugated steel pipe);

5.1.2. Type of metallic coating (zinc, aluminum-coated Type 2, aluminum-coated Type 1, 55 Al-Zn alloy, Zn-5 Al-MM alloy, or zinc and aramid fiber composite coating) (Section 6.1);

5.1.2.1. For Zn-5 Al-MM alloy, class of coating structure (Class A, minimized, etc.) (Section 4.3);

5.1.3. AASHTO designation and date of issue (M 36-____);

5.1.4. Type of pipe (Section 4.1);

5.1.5. Diameter of circular pipe (Table 6), or span and rise of pipe-arch section (Tables 8, 9, 10, or 11);

5.1.6. Length, either total length or length of each piece and number of pieces;

5.1.7. Description of corrugations (Section 7.2);

5.1.8. Sheet thickness (Section 8.1.2);

5.1.9. For Type I and Type II pipe, the pipe fabrication method, whether with annular corrugations or helical corrugations (Section 7.1.1) (Note 2);

Note 2—Pipe with annular corrugations with spot welded or riveted seams is designed by different criteria compared to pipe with helical corrugations. Pipe with annular corrugations must consider seam strength. Therefore, consideration of the method of fabrication is important when pipe is installed under certain conditions of loading.

5.1.10. When zinc and aramid fiber composite coated sheet is used for fabrication of pipe, the type of asphalt coating (Sections 1.1.1 and 8.5);

Note 3—See M 190 for additional information appropriate to bituminous post-coatings on pipe.

5.1.11. Joining systems, including the type of joint from Section 9.1 and gasket, if required (If no joining system is specified, the fabricator shall select the system.);

5.1.12. For Type III pipe, class of perforations, if other than Class 1 (Section 8.3.2);

5.1.13. Certification, if required (Section 14.1); and

5.1.14. Special requirements.

6. MATERIALS

6.1. *Steel Sheet for Pipe*—All pipe fabricated under this specification shall be formed from zinc-coated sheet conforming to M 218, or aluminum-coated Type 2 sheet conforming to M 274M, 55 percent aluminum-zinc alloy-coated sheet conforming to M 289 or zinc-5 percent aluminum-mischmetal alloy-coated sheet conforming to ASTM A 929/A 929M or aluminum-coated Type 1 conforming to ASTM A 929/A 929M. If the type of metallic coating is not stated in the order, zinc-coated sheet conforming to M 218 shall be used. All pipe furnished on the order shall have the same metallic coating, unless otherwise specified.

6.2. *Steel Sheet for Coupling Bands*—The sheet used in fabricating coupling bands shall have the same coating and shall conform to the same specification listed in Section 6.1 as that used for fabrication of the pipe furnished under the order.

axis of the pipe for pipe diameters larger than 525 mm, and not less than 45 degrees from the axis for pipe diameters of 525 mm and smaller.

7.2.1. For Type I and IA pipe, corrugations shall form smooth continuous curves and tangents. The dimensions of the corrugations shall be in accordance with Table 1 for the size indicated in the order, except if the depth measurement of one or more corrugations is less than the minimum depth in Table 1, the depth of all corrugations between adjacent seams shall be measured and the values of Table 2 for minimum average depth and minimum corrugation depth shall apply.

Table 1—Corrugation Requirements for Type I, IA, II, IIA, and III Pipe

Nominal Size, mm	Maximum Pitch,[a] mm	Minimum Depth,[b] mm	Inside Radius[c] Nominal, mm	Inside Radius[c] Minimum, mm
38 by 6.5[d]	48	6.0	7	6.5
68 by 13	73	12	17	12
75 by 25	83	24	14	12
125 by 25	135	24	40	36

[a] Pitch is measured from crest to crest of corrugation, at 90 degrees to the direction of corrugation.

[b] Depth is measured as the vertical distance from a straightedge resting on the corrugation crests parallel to the axis of the pipe to the bottom of the intervening valley. If the depth measurement of one or more corrugations is less than the value indicated herein, the depth of all corrugations between seams shall be measured, and the requirements of Table 2 shall be applied. (See Section 7.2.1.)

[c] Minimum inside radius requirement does not apply to a corrugation containing a helical lock seam.

[d] The corrugation size of 38 by 6.5 mm is available only in helically corrugated pipe.

Table 2—Referee Requirements for Corrugation Depth[a]

Nominal Size, mm	Diameter, mm	Minimum Average Depth, mm	Minimum Corrugation Depth, mm
38 by 6.5	All	6.1	5
68 by 13	300 through 525	12.1	10
68 by 13	Over 525	12.4	11
75 by 25	All	24.9	23
125 by 25	All	24.9	23

[a] See Section 7.2.1 for application of Table 2.

Note 4—Inspection frequently consists of measurement of the depth of one or a few corrugations. If such measurement indicates insufficient depth, application of the requirements in Table 2 provides for acceptance where greater depth of some corrugations compensates for lack of depth of others. These measurements would normally be made at one location between seams on a length of pipe.

7.2.2. For Type IR pipe, the corrugations shall be essentially rectangular ribs projecting outward from the pipe wall. The dimensions and spacing of the ribs shall be in accordance with Table 3 for the size indicated on the order. For the 292-mm rib spacing, if the sheet between the ribs does not include a lock seam, a stiffener shall be included midway between the ribs. This stiffener shall have a nominal radius of 6.4 mm and a minimum height of 5.1 mm toward the outside of the pipe.

Table 3—Rib Requirements for Types IR and HR Pipe

Nominal Size, mm	Rib Width,[a] Min, mm	Rib Depth,[b] Min, mm	Rib Spacing,[c] Min, mm	Bottom Outside Radius, Min, mm	Bottom Outside Radius,[d] Max Avg, mm	Top Outside Radius, Min, mm	Top Outside Radius,[d] Max Avg, mm
19 by 19 by 190	17	19	197	2.5	6.0	$2.5 + t$	$6.0 + t$
19 by 25 by 216	17	24	222	2.5	6.0	$2.5 + t$	$6.0 + t$
19 by 25 by 292	17	24	298	2.5	6.0	$2.5 + t$	$6.0 + t$

[a] Width is a dimension of the inside of the rib, but is measured on the outside of the pipe (outside of the rib) and shall meet or exceed the stated minimum width plus two times the wall thicknesses, that is, $2t + 17$ mm.

[b] Depth is an average of three ribs within one sheet width.

[c] Spacing is an average of three adjacent rib spacings for 19 by 19 by 190 pipe and two adjacent rib spacings for 19 by 25 by 292 pipe measured center-to-center of the ribs, at 90 degrees to the direction of the ribs.

[d] The average of the two top rib radii and of the two bottom rib radii shall be within the minimum and maximum tolerances. The term "outside" refers to the outside surface of the pipe.

Note 5—The nominal dimensions and properties for smooth corrugations and for ribs are given in AASHTO's *Standard Specifications for Highway Bridges*, Division I, Section 12, and in ASTM A 796.

7.3. *Riveted Seams*—The longitudinal seams shall be staggered to the extent that no more than three thicknesses of sheet are fastened by any rivet. Pipe to be reformed into pipe-arch shape shall have seams meeting the longitudinal seam requirement of Section 8.2.2. (See also Note 6.)

Note 6—Fabrication of pipe without longitudinal seams in 120 degrees of arc, so that the pipe may be installed without longitudinal seams in the invert, is subject to negotiation between the purchaser and fabricator.

7.3.1. The size of rivets, number per corrugation, and width of lap at the longitudinal seam shall be as stated in Table 4, depending on sheet thickness, corrugation size, and diameter of pipe. For pipe with 25-mm deep corrugations, M12 diameter bolts and nuts may be used in lieu of rivets on a one-for-one replacement ratio. Circumferential seams shall be riveted using rivets of the same size as for longitudinal seams and shall have a maximum rivet spacing of 150 mm, measured on centers, except that six rivets will be sufficient in 300-mm diameter pipe.

Table 4—Riveted and Spot-Welded Longitudinal Seams

Specified Sheet Thickness, mm	68 by 13 mm[a,b]	75 by 25 mm[c,d]	125 by 25 mm[e,d]
	Rivet or Spot-Weld Diameters, Min, mm		
1.32	8.0	—	—
1.63	8.0	9.5	9.5
2.01	8.0	9.5	9.5
2.77	9.5	11.0	11.0
3.51	9.5	11.0	11.0
4.27	9.5	11.0	11.0

[a] One rivet or spot weld each valley for pipe diameters 900 mm and smaller. Two rivets or spot welds each valley for pipe diameters 1050 mm and larger.

[b] Minimum width of lap: 38 mm for pipe diameters 900 mm and smaller, and 75 mm for pipe diameters 1050 mm and larger.

[c] Two rivets or spot welds each valley for all pipe diameters.

[d] Minimum width of lap: 75 mm for pipe of all diameters.

[e] Two rivets or spot welds each crest and valley for all pipe diameters.

7.3.2. All rivets shall be driven cold in such a manner that the sheets shall be drawn tightly together throughout the entire lap. The center of a rivet shall be no closer than twice its diameter from the edge of the sheet. All rivets shall have neat, workmanlike, and full hemispherical heads or heads of a form acceptable to the purchaser, shall be driven without bending, and shall completely fill the hole.

7.4. *Resistance Spot Welded Seams*—The longitudinal seams shall be staggered to the extent that no more than three thicknesses of sheet are fastened by any spot weld. Pipe to be reformed into pipe-arch shape shall also meet the longitudinal seam requirement of Section 8.2.2 (Note 6).

7.4.1. The size of spot welds, number per corrugation, and width of lap at the longitudinal seam shall be as stated in Table 4, depending on sheet thickness, corrugation size, and diameter of pipe. Circumferential seams shall be welded using spot welds of the same size as for longitudinal seams and shall have a maximum weld spacing of 150 mm, except that six welds will be sufficient in 300-mm diameter pipe.

7.4.2. All spot welds shall be made in such a manner that the sheets will be drawn tightly together throughout the lap. The outside edge of each spot weld shall be at least 6.5 mm from the edge of the sheet. The welding shall be performed in such a manner that the exterior surfaces of 90 percent or more of the spot welds on a length of pipe shall show no evidence of melting or burning of the base metal, and the base metal shall not be exposed when the area adjacent to the electrode contact surface area is wire brushed. Discoloration of the spot weld surfaces will not be cause for rejection.

7.4.3. Welding equipment shall be qualified before use, and the qualification shall be verified before each work shift and when changing sheet thickness, all as described in Appendix A1. If use of the equipment at the approved machine settings fails to produce satisfactory welds, fabrication shall be stopped until adjustments are made and the equipment is requalified.

7.5. *Helical Lock Seams*—The lock seam for Type I pipe shall be formed in the tangent element of the corrugation profile with its center near the neutral axis of the corrugation profile. The lock seam for Type IA pipe shall be in the valley of the corrugation, shall be spaced not more than 760 mm apart, and shall be formed from both the liner and the shell in the same general manner as Type I helical lock seam pipe. The lock seam for Type IR pipe shall be formed in the flat zone of the pipe wall, midway between two ribs.

7.5.1. The edges of the sheets within the cross section of the lock seam shall lap at least 4.0 mm for pipe 250 mm or less in diameter and at least 7.9 mm for pipe greater than 250 mm in diameter, with an occasional tolerance of −10 percent of lap width allowable. The lapped surfaces shall be in tight contact. The profile of the sheet shall include a retaining offset adjacent to the 180-degree fold (as described in T 249M) of one sheet thickness on one side of the lock seam, or one-half sheet thickness on both sides of the lock seam, at the fabricator's option. There shall be no visible cracks in the metal, loss of metal-to-metal contact, or excessive angularity on the interior of the 180-degree fold of metal at the completion of forming the lock seam.

7.5.2. Specimens cut from production pipe normal to and across the lock seam shall develop the tensile strength as provided in Table 5, when tested according to T 249. For Type IA pipe, the lock seam strength shall be as tabulated based on the thickness of the corrugated shell.

Table 5—Lock Seam Tensile Strength

Specified Sheet Thickness,[a] mm	Lock Seam Tensile Strength, per Unit Width, Min, kN/m
1.02	30
1.32	42
1.63	60
2.01	91
2.77	122
3.50	154
4.27	210

[a] For Type IA pipe, the thickness shall be that of the corrugated shell.

7.5.3. When the ends of helically corrugated lock seam pipe have been rerolled to form annular corrugations, either with or without a flanged end finish, the lock seam in the rerolled end shall not contain any visible cracks in the base metal and the tensile strength of the lock seam shall be not less than 60 percent of that required in Section 7.5.2.

7.6. *Helical Continuous Welded Seams*—The seam shall be parallel to the corrugations and shall have a continuous weld extending from end to end of each length of pipe. Welding shall be done utilizing ultra-high frequency resistance equipment. Seams shall be welded in such a manner that they will develop the full strength of the pipe and not affect shape or nominal diameter of the pipe. Welded seams shall be controlled such that the combined width of weld and adjacent coating burned by welding does not exceed three times the metal thickness. Damage outside this width shall be repaired as required in Section 11. The fabricator shall certify that the welds have been tested and found satisfactory.

7.6.1. Continuous welded seams shall be tested in accordance with the cup test procedure (Section 3) of T 241. The welded seam shall be acceptable if the sum of the length of cracks or other defects on either side of the cup does not exceed 6.5 mm, basing the result on the second test if the first shows greater defects. The provisions of the referee test method of Section 4 of T 241 shall be applicable in the event of disagreement between the purchaser and the fabricator.

7.6.1.1. Tests of continuous welded seams shall be made as follows:

7.6.1.2. Pipe lengths of 7.3 m or less shall be tested on one end of each length, normally the trailing end.

7.6.1.3. If a length of pipe having a diameter greater than 1200 mm and length of 7.3 m or less is rejected, the following length of pipe produced shall be tested on both ends. If the test on either end fails, this entire length shall also be rejected.

7.6.1.4. Pipe lengths greater than 7.3 m shall be tested on each end of each length of pipe. If either end fails, the entire length shall be rejected.

7.6.2. The requirement for conducting quality control tests in accordance with Section 7.6.1 shall not apply for pipe in which the ends have been rerolled to form annular corrugations. The manufacturer shall maintain visual evaluation of the quality of the weld after rerolling and any indication of weld or base metal failure will be cause for rejection of the pipe.

7.6.3. Any indication of cracks, skips, or deficient welds found through visual inspection will be cause for rejection unless repaired. It is the option of the fabricator to remove the defective portion of the length of pipe or to manually repair defects in the automatically welded seam. Altered or repaired pipe shall meet the applicable requirements of Section 7.6. Where a manual repair occurs within 400 mm of the end of the length of pipe, a test shall be conducted on both the manually repaired section and on the immediately adjacent automatically welded section. If either test results in failure under the criterion of Section 7.6.1, the length of pipe shall be rejected.

7.7. *End Finish:*

7.7.1. To facilitate field jointing, the ends of the individual pipe sections with helical corrugations or ribs may be rerolled to form annular corrugations extending at least two corrugations from the pipe end, or to form an upturned flange meeting the requirements in Section 7.7.2, or both. The diameter of ends shall not exceed that of the pipe barrel by more than the depth of the corrugation. All types of pipe ends, whether rerolled or not, shall be matched in a joint such that the maximum difference in the diameter of abutting pipe ends is 13 mm.

7.7.1.1. When pipe with any size helical corrugation or rib is rerolled to form annular corrugations in the ends, the usual size of the annular corrugation is 68 by 13 mm.

7.7.2. If a flanged finish is used on the ends of individual pipe sections to facilitate field jointing, the flange shall be uniform in width, be not less than 13 mm wide, and shall be square to the longitudinal axis of the pipe.

7.7.3. The ends of all pipe which will form the inlet and outlet of culverts, fabricated of sheets having nominal thicknesses of 2.01 mm and less, shall be reinforced in a manner approved by the purchaser, when specified.

8. PIPE REQUIREMENTS

8.1. *Type I, Type IA, and Type IR Pipe:*

8.1.1. *Pipe Dimensions*—The nominal diameter of the pipe shall be as stated in the order, selected from the size listed in Table 6. The size of corrugations that are standard for each size of pipe are also shown in Table 6. The average inside diameter of circular pipe and pipe to be reformed into pipe-arches shall not vary more than 1 percent or 13 mm, whichever is greater, from the nominal diameter when measured on the inside crest of the corrugations for Type I pipe, or the inside liner or surface for Type IA or Type IR pipe, respectively. Alternately, for pipe having annular corrugations, conformance with the inside diameter requirement may be determined by measuring the outside circumference, for which minimum values are given in Table 6.

Note 7—The outside circumference of helically corrugated pipe is influenced by the corrugation size and the angle of the corrugation, affecting the number of corrugations crossed; therefore, no minimum measurement can be specified.

8.1.2. *Sheet Thickness*—Sheet thickness shall be specified by the purchaser from the specified sheet thicknesses listed in Table 7 (Notes 8 and 9). For Type IA pipe, the thickness of both the shell and the liner shall be given; the thickness of the corrugated shell shall not be less than 60 percent of the thickness of the equivalent Type I pipe; the liner shall have a nominal thickness of at least 1.02 mm; and the sum of the specified thicknesses of shell and liner shall equal or exceed the specified thickness of an equivalent pipe of identical corrugations as the shell according to the design criteria in AASHTO's *Standard Specifications for Highway Bridges*.

Table 6—Pipe Sizes

Nominal Inside Diameter, mm	Corrugation Sizes[a]				Ribbed Pipe			Minimum Outside Circumference,[c] mm
	38 by 6.5 mm	68 by 13 mm	75 by 25 mm	125 by 25 mm	19 by 19 by 190 mm[b]	19 by 25 by 292 mm	19 by 25 by 216 mm	
100	X							264
150	X							441
200	X							598
250	X							755
300	X	X						912
375	X	X						1148
450	X	X			X	X	X	1383
500		X			X	X	X	1620
600		X			X	X	X	1854
675		X			X	X	X	2091
750		X			X	X	X	2483
825		X		X	X	X	X	2561
900		X	X	X	X	X	X	2797
1050		X	X	X	X	X	X	3269
1200		X	X	X	X	X	X	3739
1350		X	X	X	X	X	X	4209
1500		X	X	X	X	X	X	4675
1650		X	X	X	X	X	X	5142
1800		X	X	X	X	X	X	5609
1950		X	X	X	X	X	X	6075
2100		X	X	X	X	X	X	6542
2250			X	X	X	X	X	7008
2400			X	X	X	X	X	7475
2550			X	X	X	X	X	7941
2700			X	X	X	X	X	8408
2850			X	X	X		X	8874
3000			X	X	X		X	9341
3150			X	X			X	9807
3300			X	X			X	10274
3450			X	X			X	10740
3600			X	X			X	11207

[a] An "X" indicates standard corrugation sizes for each nominal diameter of pipe.

[b] Rib sizes 19 by 19 by 190 mm and 19 by 25 by 292 mm.

[c] Measured in valley of annular corrugations. Not applicable to helically corrugated pipe.

Table 7—Thicknesses of Metallic-Coated Steel Sheet[a]

Specified Thickness, mm	M 218, Zinc Coated	M 274 AlT2, Aluminum Coated	M 289, 55 Percent Aluminum-Zinc Alloy Coated	A 885, Zinc and Aramid Fiber Coated	A 929M Zn-5 Al-MM, Alloy Coated	A 929M AlT1, Aluminum Coated
	Specification Designation					
1.02	X		X		X	
1.32	X	X	X		X	X
1.63	X	X	X	X	X	X
2.01	X	X	X	X	X	X
2.77	X	X	X	X	X	X
3.51	X	X	X	X	X	X
4.27	X			X	X	

[a] An "X" indicates sheet thickness included in the applicable specification for coating types listed.

Note 8—The sheet thicknesses indicated in Table 7 are the thicknesses listed as available in M 218, M 274, M 289, ASTM A 885, and ASTM A 929M for zinc and aramid fiber composite coated sheet.

Note 9—The purchaser should determine the required thickness for each of the types of pipe described in Section 4.1.1 through Section 4.1.6 according to the design criteria in AASHTO's *Standard Specifications for Highway Bridges*, Division 1, Section 12, or other appropriate guidelines.

8.1.3. When specified by the purchaser, the finished pipe shall be factory elongated to the extent specified. The elongation shall be accomplished by the use of a mechanical apparatus, which will produce a uniform deformation throughout the length of the section.

8.2. *Type II, IIA, and IIR Pipe*:

8.2.1. *Pipe-Arch Dimensions*—Pipe furnished as Type II, IIA, or IIR shall be made from Type I, IA, or IR pipe respectively, and shall be reformed to provide a pipe-arch shape. All applicable requirements for Types I, IA, and IR pipe shall be met by finished Types II, IIA, and IIR pipe, respectively. Pipe-arches shall conform to the dimensional requirements of Tables 8, 9, 10, or 11. All dimensions shall be measured from the inside crests of corrugations for Type II pipe or from the inside liner or surface for Types IIA or IIR, respectively.

8.2.2. *Longitudinal Seams*—Longitudinal seams of riveted or spot-welded pipe-arches shall not be placed in the corner radius.

8.2.3. Reforming Type IR into Type IIR pipe shall be done in such a manner as to avoid damage to the external ribs.

8.3. *Type III Pipe*:

8.3.1. Type III pipe shall have a full circular cross-section and shall conform to the requirements for Type I pipe and, in addition, shall contain perforations conforming to one of the classes described in Section 8.3.2.

Table 8—(M 36) Pipe-Arch Requirements 68 by 13 mm Corrugations

Pipe Arch Size, mm	Equivalent Dia, mm	Span,[a] mm	Rise,[a] mm	Minimum Corner Radius, mm	Maximum B,[b] mm
430 by 330	375	430	330	75	135
530 by 380	450	530	380	75	155
610 by 460	525	610	460	75	185
710 by 510	600	710	510	75	205
780 by 560	675	780	560	75	225
885 by 610	750	870	610	75	240
970 by 690	825	970	690	75	255
1060 by 740	900	1060	740	90	265
1240 by 840	1050	1240	840	100	290
1440 by 970	1200	1440	970	130	345
1620 by 1100	1350	1620	1100	155	380
1800 by 1200	1500	1800	1200	180	420
1950 by 1320	1650	1950	1320	205	460
2100 by 1450	1800	2100	1450	230	510

[a] A tolerance of 25 mm or two percent of equivalent diameter, whichever is greater, will be permissible in span and rise.

[b] B is defined as the vertical dimension from a horizontal line across the widest portion of the arch to the lowest portion of the base.

Table 9—(M 36) Pipe-Arch Requirements 75 by 25 mm or 125 by 25 mm Corrugations

Pipe Arch Size, mm	Equivalent Dia, mm	Span,[a] mm	Rise,[a] mm	Minimum Corner Radius, mm
1010 by 790	900	1010 – 45	790 + 45	130
1160 by 920	1050	1160 – 55	920 + 55	155
1340 by 1050	1200	1340 – 60	1050 + 60	180
1520 by 1350	1350	1520 – 70	1170 + 70	205
1670 by 1300	1500	1670 – 75	1300 + 75	230
1850 by 1400	1650	1850 – 85	1400 + 85	305
2050 by 1500	1800	2050 – 95	1500 + 95	355
2200 by 1620	1950	2200 – 110	1620 + 110	355
2400 by 1720	2100	2400 – 120	1720 + 120	410
2600 by 1820	2250	2600 – 130	1820 + 130	410
2840 by 1920	2400	2840 – 145	1920 + 145	460
2970 by 2020	2550	2970 – 150	2020 + 150	480
3240 by 2120	2700	3240 – 165	2120 + 165	480
3470 by 2220	2850	3470 – 175	2220 + 175	480
3600 by 2320	3000	3600 – 180	2320 + 180	480

[a] Negative and positive numbers listed with span and rise dimensions are negative and positive tolerances; no tolerance in opposite direction.

Table 10—Pipe-Arch Requirements, 19 by 19 by 190 mm Rib Corrugations

Pipe Arch Size, mm	Equivalent Diameter, mm	Span,[a] mm	Rise,[a] mm	Minimum Corner Radius, mm
500 by 410	450	500 – 25	410 + 25	130
580 by 490	525	550 – 25	490 + 25	130
680 by 540	600	680 – 40	540 + 40	130
750 by 620	675	750 – 40	620 + 40	130
830 by 670	750	830 – 40	670 + 40	130
900 by 750	825	900 – 45	750 + 45	130
1010 by 790	900	1010 – 45	790 + 45	130
1160 by 920	1050	1160 – 55	920 + 55	155
1340 by 1050	1200	1340 – 60	1050 + 60	180
1520 by 1170	1350	1520 – 70	1170 + 70	205
1670 by 1300	1500	1670 – 75	1300 + 75	230
1850 by 1400	1650	1850 – 85	1400 + 85	305
2050 by 1500	1800	2050 – 95	1500 + 95	355
2200 by 1620	1950	2200 – 100	1620 + 100	355
2400 by 1720	2100	2400 – 105	1720 + 105	410
2600 by 1820	2250	2600 – 115	1820 + 115	410
2840 by 1920	2400	2840 – 120	1850 + 120	450
2920 by 1980	2550	2920 – 130	1980 + 130	450

[a] Negative and positive numbers listed with span and rise dimensions are negative and positive tolerances; no tolerance in opposite direction.

Table 11—Pipe Arch Requirements, 19 by 25 by 292 mm Rib Corrugation

Pipe Arch Size, mm	Equivalent Diameter, mm	Span,[a] mm	Rise,[a] mm	Minimum Corner Radius, mm
500 by 410	450	500 – 25	410 + 25	130
580 by 490	525	550 – 25	490 + 25	130
680 by 540	600	680 – 40	540 + 40	130
750 by 620	675	750 – 40	620 + 40	130
830 by 670	750	830 – 40	670 + 40	130
900 by 750	825	900 – 45	750 + 45	130
1010 by 790	900	1010 – 45	790 + 45	130
1160 by 920	1050	1160 – 55	920 + 55	155
1340 by 1050	1200	1340 – 60	1050 + 60	180
1520 by 1170	1350	1520 – 70	1170 + 70	205
1670 by 1300	1500	1670 – 75	1300 + 75	230
1850 by 1400	1650	1850 – 85	1400 + 85	305
2050 by 1500	1800	2050 – 95	1500 + 95	355

8.3.2. *Perforations*—The perforations shall conform to the requirements for Class 1, unless otherwise specified in the order. Class 1 perforations are for pipe intended to be used for subsurface drainage. Class 2 and 3 perforations are for pipe intended to be used for subsurface disposal of water, but pipe containing Class 2 and 3 perforations may also be used for subsurface drainage.

8.3.2.1. *Class 1 Perforations*—The perforations shall be approximately circular and cleanly cut; shall have nominal diameters of not less than 4.8 mm nor greater than 9.5 mm; and shall be arranged in rows parallel to the axis of the pipe. The perforations shall be located on the inside crests or along the neutral axis of the corrugations, with one perforation in each row for each corrugation. Pipe connected by couplings or bands may be unperforated within 100 mm of each end of each length of pipe. The rows of perforations shall be arranged in two equal groups placed symmetrically on either side of a lower unperforated segment corresponding to the flow line of the pipe. The spacing of the rows shall be uniform. The distance between the centerlines of rows shall be not less than 25 mm. The minimum number of longitudinal rows of perforations, the maximum heights of the centerlines of the uppermost rows above the bottom of the invert, and the inside chord lengths of the unperforated segments illustrated in Figure 1 shall be as specified in Table 12.

Note 10—Pipe with Class 1 perforations is generally available in diameters from 100 to 525 mm inclusive, although perforated pipe in larger sizes may be obtained.

8.3.2.2. *Class 2 Perforations*—The perforations shall be circular holes with nominal diameters of 8.0 to 9.5 mm, or slots with nominal widths of 4.8 to 8.0 mm and not to exceed 13 mm in length. The perforations shall be uniformly spaced around the full periphery of the pipe. The perforations shall provide an opening area of not less than 230 square centimeters per square meter of pipe surface based on nominal diameter and length of pipe.

Note 11—323 perforations, 9.5-mm diameter, per square meter satisfies this requirement.

8.3.2.3. *Class 3 Perforations*—The perforations shall be slots with a width of 2.5 ± 1.0 mm and length of 25 ± 6.5 mm, spaced 45 to 65 mm on centers around the circumference and staggered on the outside crests of the corrugations of the pipe. No metal shall be removed in making the slot. Slots shall be made from the inside of the pipe.

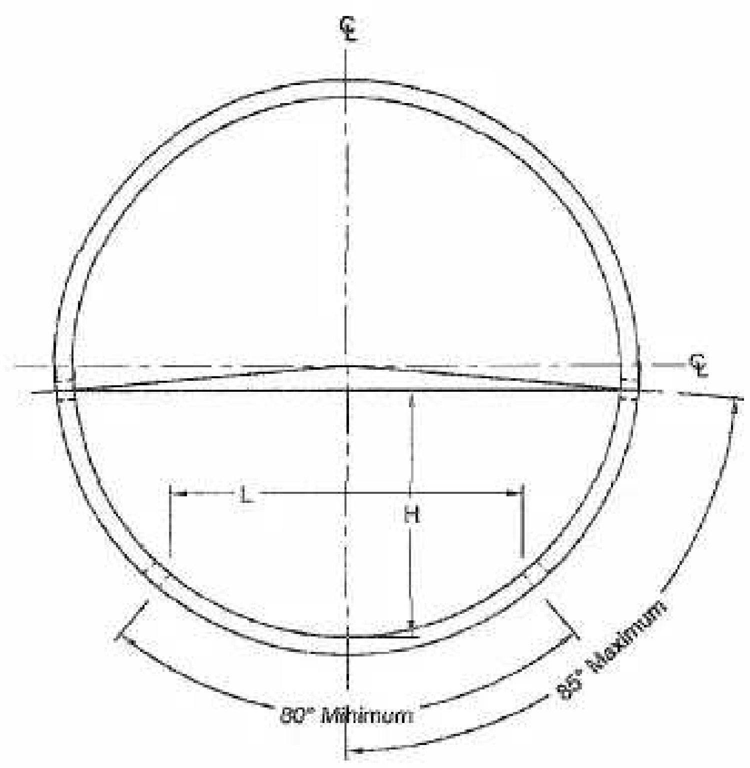

Figure 1—Requirements for Perforations

Table 12—Rows of Perforations, Height *H* of the Centerline of the Uppermost Rows Above the Invert, and Chord Length *L* of Unperforated Segment, for Class 1 Perforations

Internal Diameter of Pipe, mm	Rows of Perforations[a]	*H*, Max,[b] mm	*L*, Min,[b] mm
100	2	46	64
150	4	69	96
200	4	92	128
250	4	115	160
300	6[c]	138	192
375	6[c]	172	240
450	6[c]	207	288
525	6	241	336
600 and larger	8	[d]	[d]

[a] Minimum numbers of rows. A greater number of rows for increased inlet area shall be subject to agreement between purchaser and fabricator. Note that the number of perforations per unit length in each row (and inlet area) is dependent on the corrugation pitch.

[b] See Figure 1 for location of dimensions *H* and *L*.

[c] Minimum of 4 rows permitted in pipe with 38 by 6.5 mm corrugations.

[d] $H(\text{max}) = 0.46D$, $L(\text{min}) = 0.64D$, where D = internal diameter of pipe, mm or in. as appropriate.

8.4. *Type IIIA Pipe*:

8.4.1. Type IIIA pipe shall be fabricated of an unperforated semicircular bottom section with a top shield of corrugated steel, both of nominal 1.32-mm thickness or greater. The smooth semi-circular bottom section shall be approximately 120 mm in diameter and shall have a continuous lip extending outward along each side; the corrugated top shield shall be approximately 160 mm wide including a 19-mm sloping overhang on each side and shall be secured to the lip of the bottom section by integral tabs spaced at about 90 mm center-to-center. The top shield shall have corrugations approximately 22 mm center-to-center and approximately 8.0-mm depth.

8.5. *Pipe Fabricated from Zinc and Aramid Fiber Composite Coated Sheet*—Pipe which has been fabricated from zinc and aramid fiber composite coated sheet shall be coated with asphalt as described in M 190, Type A, Fully Bituminous Coated. If full or partial smooth lining is desired, it shall be specified by the purchaser. (See Section 1.1.1 and M 190.)

9. JOINING SYSTEMS

9.1. *Types of Joining Systems*—Joining systems shall be of the following types, depending upon the configuration of the steel band joining the pipe together. If required, the joining system shall incorporate a flat, O-ring, or profile gasket. The corrugations at the ends of pipe sections being joined shall conform to one of the corrugations detailed in ASTM A 796/A 796M.

Note 12—Bands are classified according to their ability to resist shear, moment, and tensile forces as described in AASHTO's *Standard Specifications for Highway Bridges*, Division II, Section 26, and identified as "standard joints" and "special joints." The first five types of bands listed in Section 9.1 and meeting the requirements of Section 9.2 are expected to meet the requirements for "standard joints." Some may also be able to meet the requirements for "special joints," but such capability should be determined by analysis or test.

9.1.1. *Corrugated Bands*—Bands with either annular or helical corrugations. The band corrugation shall match that of the pipe sections being joined or the annular rerolled ends of those pipe section.

9.1.2. *Partially Corrugated Bands*—Flat bands with a minimum of one corrugation formed along each circumferential edge of the band. These bands are intended for use with helically corrugated pipe with its ends rerolled to a 68 mm by 13 mm corrugation.

9.1.3. *Bands with Projections*—Flat bands with projections such as dimples, are used to join pipe with either helical or annular corrugations. The bands shall be formed with the projections in annular rows with one projection for each corrugation of helical pipe engaged by the band. Bands 265 or 300 mm wide shall have two annular rows of projections, and bands 415 or 560 mm wide shall have four annular rows of projections.

9.1.4. *Channel Bands*—Channel bands that incorporate a connector formed into a channel (hat) shape, shall be used only with pipe having upturned flanges on the pipe ends. Channel bands shall conform with the requirements of Section 9.3.3.

9.1.5. *Flat Bands*—When specified by the purchaser, flat bands shall be used on pipe with helically corrugated ends, annular corrugated pipe, or helical pipe on which the ends have been rerolled to form annular corrugations.

9.1.6. *Sleeve Couplers*—When specified by the purchaser, the joining system shall incorporate a push-on type coupler designed to properly interface with the pipes being joined. Sleeve couplers generally do not have any external device for tightening around the pipe. Sleeve couplers shall provide a centering device so the coupler laps equally on both pipes being joined. Sleeve couplers for pipe diameters less than 300 mm shall have a minimum stab depth of 75 mm. The minimum stab depth for 300-mm through 1050-mm diameters shall be 150 mm. When sleeve couplers are used with pipes other than Type III or Type IIIA, pipe with annular corrugations or rerolled ends shall be used. Sleeve couplers are not intended for pipe diameters larger than 1050 mm.

9.1.7. *Bell and Spigot*—Bell and spigot configurations incorporate an integral bell that is permanently installed at the factory to one end of the pipe, while the other end of the pipe serves as a spigot. The bell shall be affixed to the pipe by welding or with mechanical fasteners. The steel in the bell shall meet the thickness requirements of Section 9.3.1. The bell and spigot configuration shall be classified in accordance with Section 9.2. The spigot end of the pipe shall be re-rolled or provide annular corrugations to allow placement of a gasket if required. The bell shall provide a minimum stab depth of 150 mm, or 8 percent of the pipe diameter, which ever is greater.

9.1.8. *Special Design*—Other joining systems that meet the requirements of Sections 9.3, 9.4 (if required), and 9.5 (if required), and are specified by the purchaser, shall be used on the project for which they are specified.

9.2. *Joining Systems, Significance and Use*—Joining systems are classified as standard or gasketed, based on the ability of the system to control infiltration and/or exfiltration. These classifications of standard and gasketed systems are covered in Sections 9.2.1 and 9.2.2. When site conditions near the pipe are such that movement or differential settlement is possible at the joint, the joining system shall have structural requirements that conform with Section 9.4.

9.2.1. *Standard Systems*:

9.2.1.1. Standard joining systems for corrugated steel pipe are intended to control the infiltration of soil into the pipe. Standard joining systems are used in the majority of CSP installations. The gradation and plasticity of the backfill materials around the pipe are important factors in the selection of a joining system. The greater the concentration of very fine particles, such as silts which pass the 75-µm sieve, the greater the possibility of soil infiltration. Additionally, when the water table is above the pipe or where the backfill is otherwise saturated and the flow level in the pipe varies rapidly, soil infiltration is more likely to occur.

9.2.1.2. External bands conforming with Sections 9.1.1–9.1.3, or 9.1.5, when used with annular pipe or with helical pipe with rerolled ends, generally provide adequate control of infiltration of soil particles. Properly installed, these bands provide continuous metal-to-metal contact with the periphery of the annular portion of the pipe. Where these bands do not provide adequate soil infiltration control, a geotextile wrap around the exterior of the joining system and the adjacent pipe will inhibit the movement of silt and larger soil particles into the pipe.

9.2.2. *Gasketed Systems*:

9.2.2.1. Gasketed joining systems are used to limit the flow of water from the pipe interior to the backfill, to limit the flow of groundwater into the pipe, and where necessary, to provide further control of soil particle infiltration. Joining systems such as those described in Sections 9.1.4, 9.1.6, and 9.1.7 require gaskets to adequately control soil particle infiltration, except where the pipe is used in underdrain applications. Gaskets used in all joining systems shall conform to the requirements of Sections 6.7 and 9.5.

9.2.2.2. Sites where excessive water infiltration is expected or where the pipe will carry hazardous pollutants require a gasketed joining system. In these cases, the joining system shall be tested to establish a leakage rate not to exceed 18.5 L per millimeter of diameter per kilometer per day with no pressure applied to the pipe or joining system. The joining system shall be tested at the pipe fabricator's plant or a laboratory, with the pipe in straight alignment. The test shall be witnessed and certified by an approved laboratory.

9.3. *Requirements*—Joining systems shall be fabricated in a manner that ensures that the band or coupler extends over each pipe section an equal length. The joining system shall be fabricated in such a way that proper installation will result in performance conforming with Sections 9.2.1 or 9.2.2 as required for the project.

9.3.1. *Band, Sleeve, or Bell Thickness and Width*—The band, sleeve coupler, or bell portion of the joining system shall be sufficiently strong to resist the forces to which it is subjected. Table 13 provides minimum steel thickness requirements for bands, sleeves, or bells based on the steel thickness of the pipes being connected. Table 13 does not apply to channel (hat) bands, which are covered under Section 9.3.3. The width of coupling bands (Sections 9.1.1–9.1.3 and 9.1.5) shall be equal to or greater than the minimum widths shown in Tables 14 and 15.

9.3.2. *Band Connectors*—The bands shall be connected in a manner approved by the purchaser with hardware that has been suitably galvanized to provide durability. This hardware includes angles and integrally or separately formed and attached flanges that will be connected together with galvanized or cadmium-plated bolts, bars and straps, wedge locks, and straps or lugs. Bands shall be connected with the bolts in accordance with Table 16.

9.3.3. *Channel Bands*—Pipe sections provided with flanges on the ends will be connected by interlocking the flanges of two pipes with a channel (hat) band or other band incorporating an interlocking channel, not less than 19 mm in width. The depth of the channel shall be not less than 13 mm. The channel band shall have a minimum thickness of 1.62 mm.

9.3.4. *Sleeve Couplers*—Sleeve couplers for pipes less than 300 mm in diameter shall be made from steel with a minimum thickness of 1.02 mm. The steel thickness for larger sizes shall conform to Table 13. Alternatively the coupler shall be a plastic sleeve with adequate strength to maintain the in-service pipe alignment and meet the requirements of Section 9.2.1.

9.4. *Structural Properties*—Joining systems that are subject to forces created by differential soil movement or settlement require certain structural properties to withstand the applied forces. Minimum values for these structural properties are shown in Table 17. These values for a joining system are determined by either a rational analysis or a suitable physical test.

9.4.1. *Shear Strength*—The shear strength required of the joining system is expressed as a percentage of the calculated shear strength of the pipe at a typical cross section at a location other than a rerolled end.

9.4.2. *Moment Strength*—The moment strength required of a joining system is expressed as a percent of the calculated moment strength of the pipe at a typical cross section at a location other than a rerolled end.

9.4.3. *Tensile Strength*—Where pull-apart (tensile) strength is required to control disjointing in slope drains and similar applications, corrugated, partially corrugated, or channel bands shall be specified. When special requirements exist, joining systems shall provide tensile strength levels of 22 kN for 1050 mm and smaller sizes and 46 kN for larger sizes.

9.5. *Gaskets*—Where infiltration or exfiltration is a concern, as defined in Section 9.2.2, the joining system shall incorporate gaskets. Rubber gaskets shall meet the requirements of Section 6.7 and shall be flat, O-ring, or have a profile shape. Flat gaskets shall be continuous or have a lap joint and be approximately 18 mm wide and approximately 9.5 mm thick. O-ring gaskets shall be 20 mm or 22 mm in diameter for pipe diameters of 900 mm and smaller, or 22-mm diameter for larger diameters of pipe fabricated with a 13-mm deep corrugation. O-ring gaskets shall be 35 mm in diameter when used on pipes having a 25-mm or deeper corrugation. Other types of gaskets may be used, including profile gaskets, that meet the specific requirements of the joining system.

Table 13—Band, Sleeve, or Bell Thickness[a,b]

Nominal Pipe Thickness, mm	Nominal Coupling Band Thickness, Min, mm
2.77 and thinner	1.32
3.51	1.63
4.27	2.01

[a] For annular corrugated pipe or helically corrugated pipe with 68 by 13 mm annular rerolled ends.

[b] Applies to joining systems covered by Sections 9.1.1–9.1.3, 9.1.5, 9.1.7, and 9.1.6, when the coupler is 300 mm or larger and made from steel.

Note 13—Riveted and spot-welded pipe is not watertight, having small openings at the intersection of longitudinal and circumferential seams. Therefore, these types of fabrication should not be used where watertightness is a concern unless the pipe is bituminous coated or lined prior to installation.

Table 14—Band Width Requirements for Pipe with Annular Corrugated Ends[a,b]

Nominal Pipe Diameter, mm	Minimum Band Width, mm
300 to 900	175
1050 to 3600	265

[a] For annular pipe or helical pipe with 68 by 13 mm rerolled ends.

[b] Applies to joining systems covered by Sections 9.1.1–9.1.3 and 9.1.5.

Table 15—Band Width Requirements for Helically Corrugated Pipe

Nominal Corrugation, mm	Nominal Pipe Diameter, mm	Minimum Band Width, mm
38 by 6.5	100 to 450	175
68 by 13	300 to 2100	300
75 by 25	900 to 3600	350
125 by 25	900 to 3600	550

Note 1—Corrugation of band shall match that pipe.

Note 2—Band shall be centered on pipes being joined.

Note 3—Applies to joining systems covered by Sections 9.1.1–9.1.3 and 9.1.5.

Table 16—Band Connector Bolt Size

Pipe Diameter, mm	Bolt Diameter, mm
450	M10
525	M12
Type III and IIIA	M8

Table 17—Structural Properties of Joining System

	Minimum Value
Shear strength (% of barrel strength)	2
Moment strength (% of barrel strength)	5
Tensile (pull-apart) strength	none

9.6. Other types of coupling bands or fastening devices which are equally effective as those described, and which comply with the joint performance criteria of AASHTO's *Standard Specifications for Highway Bridges*, Division II, Section 26 may be used when approved by purchaser.

10. WORKMANSHIP

10.1. The complete pipe shall show careful, finished workmanship in all particulars. Pipe that has been damaged, either during fabrication or in shipping, may be rejected unless repairs are made that are satisfactory to the purchaser. Among others, the following defects shall be considered as constituting poor workmanship:

- Variation from a straight centerline;
- Elliptical shape in pipe intended to be round;
- Dents or bends in the metal;
- Metallic coating which has been bruised, broken, or otherwise damaged;
- Lack of rigidity;
- Illegible markings on the steel sheet;
- Ragged or diagonal sheared edges;
- Uneven laps in riveted or spot-welded pipe;
- Loose, unevenly lined, or unevenly spaced rivets;
- Defective spot welds or continuous welds; and
- Loosely formed lockseams.

11. REPAIR OF DAMAGED COATINGS

11.1. Pipe on which the metallic coating has been burned by welding beyond the limits provided in Sections 7.4.2 and 7.6, or has been otherwise damaged in fabricating or handling, shall be repaired. The repair shall be done so that the completed pipe shall show careful finished workmanship in all particulars. Pipe which, in the opinion of the purchaser, has not been cleaned or coated satisfactorily may be rejected. If the purchaser so elects, the repair shall be done in his presence.

11.2. The damaged area shall be repaired in conformance with ASTM A 780 (Note 14), except as described herein. The damaged area shall be cleaned to bright metal by blast cleaning, power disk sanding, or wire brushing. The cleaned area shall extend at least 13 mm into the undamaged section of the coating. The cleaned area shall be coated within 24 hours and before any rusting or soiling.

Note 14—While ASTM A 780 specifically refers to repair of damaged zinc coatings, the same procedures are applicable to repair of other metallic coatings except as described in this section.

11.3. *Zinc-Rich Paint Coating*—Zinc-rich paint, as described in the Materials section of ASTM A 780, shall be applied to a dry film thickness of at least 0.13 mm over the damaged section and surrounding cleaned area. Zinc-rich paint shall be used for repair of damage to all types of metallic coating—zinc, aluminum, and aluminum-zinc alloy.

11.4. *Metallizing Coating*—The damaged area shall be cleaned as described in Section 11.2, except it shall be cleaned to the near-white condition. The repair coating applied to the cleaned section shall have a thickness of not less than 0.13 mm over the damaged section and shall taper off to zero thickness at the edges of the cleaned undamaged section.

11.4.1. Where zinc coating is to be metallized, it shall be done with zinc wire containing not less than 99.98 percent zinc.

11.4.2. Where aluminum coating is to be metallized, it shall be done with aluminum wire containing not less than 99 percent aluminum.

11.4.3. Where 55 percent aluminum-zinc alloy coating is to be metallized, it shall be done by using the materials described in Section 11.4.1 or Section 11.4.2, or by using an alloy wire of 55 percent aluminum and 45 percent zinc by mass.

11.4.4. When Zn-5 Al-MM alloy coating is to be metallized, it shall be done using the materials described in Section 11.4.1 or by using an alloy wire of 85 percent zinc and 15 percent aluminum by weight.

11.5. Pipe on which zinc and aramid fiber composite coating is damaged by welding during fabrication of fittings, or otherwise damaged during handling or shipping, shall be repaired as described in Sections 11.2 through 11.4.

12. INSPECTION

12.1. The purchaser or his representative shall have free access to the fabricating plant for inspection, and every facility shall be extended to him for this purpose. This inspection shall include an examination of the pipe for the items in Section 10.1 and the specific requirements of this specification applicable to the type of pipe and method of fabrication.

12.2. On a random basis, samples may be taken for chemical analysis and metallic coating measurements for check purposes. These samples will be secured from fabricated pipe or from sheets or coils of the material used in fabrication of the pipe. The mass of metallic coating shall be determined in accordance with T 65M/T 65 for zinc, 55 Al-Zn alloy, and Zn-5 Al-MM alloy, and T 213M/T 213 for aluminum.

13. REJECTION

13.1. Pipe that fails to conform to the specific requirements of this specification, or that shows poor workmanship, may be rejected. This requirement applies not only to the individual pipe, but to any shipment as a whole where a substantial number of pipe are defective. If the average deficiency in length of any shipment of pipe is greater than one percent, the shipment may be rejected.

14. CERTIFICATION

14.1. When specified in the purchase order or contract, a manufacturer's or fabricator's certification, or both, shall be furnished to the purchaser stating that samples representing each lot have been tested and inspected in accordance with this specification and have been found to meet the requirements for the material described in the order. When specified in the order, a report of the test results shall be furnished.

ANNEX

(Mandatory Information)

A1. QUALIFICATION OF RESISTANCE SPOT WELDING EQUIPMENT

A1.1. *General*—Welding equipment shall be of sufficient capacity, of such design, and in such condition as to make possible the production of first-class welds. Before being permitted to perform welding on corrugated steel pipe, resistance spot welding machines and operations shall be qualified by means of the test prescribed in Section A1.2. Tests shall be performed by the fabricator's shop or by a recognized independent laboratory at no expense to the purchaser. Qualification tests performed by the fabricator's shop shall be made in the presence of the representative of the purchaser.

A1.2. *Qualification*—Perform three tension shear tests representing each thickness of sheet to be used in the fabrication of the pipe. Prepare specimens by lapping two strips of corrugated steel sheet 38 mm minimum width by 125 mm minimum length and joining them together by a single spot weld duplicating the size to be used in production. The length of lap shall be 38 mm. The longer axis of specimen shall be parallel to the direction of rolling. Test the specimens in tension to destruction in a standard calibrated testing machine. The minimum shear strength in kilonewtons as determined by this test shall be not less than that shown in Table A1.1 for nominal thickness of sheet used in the test.

Table A1.1—Shear Strength of Spot Welds

Specified Sheet Thickness, mm	Minimum Shear Strength, kN
1.63	18.2
2.01	23.1
2.77	31.1
3.51	37.8
4.27	44.5

A1.3. *Verification*—After a machine and operator have been qualified by the foregoing procedure, to insure that qualification is maintained, make three tension shear tests at the start of each work shift, and make three tension shear tests for each change in sheet thickness.

A1.4. *Machine Setting*—One copy of the approved machine setting shall be posted on the machine for use by the machine operator. No settings shall be varied, except weld phase shift and pressure, which may be varied by ±10 percent.

[1] This specification is technically identical to ASTM A 760/A 760M-01a.